Module 12
Business Finance

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION A – CASE QUESTIONS (Total: 50 marks)

Answer 1

To:	Board of Directors, Dragon Group Limited ("DGL")
From:	Financial Controller, Flying Success Limited ("FSL")
Date:	dd/mm/yyyy
Remarks:	Evaluation of external influences on FSL's new drone delivery service business in Hong Kong's low-altitude economy ("LAE")

Introduction

The purpose of this memo is to use the PESTEL model to evaluate various external influences and determine whether they provide opportunities or threats to FSL's new drone delivery service business in Hong Kong. PESTEL is a framework that evaluates the political, economic, social, technological, environmental, and legal aspects of a business and the competitive environment influencing a company.

Political influences

Political influences include government policies and licensing authorities that influence the formation and development of a company and business. The Government has been forming and developing working groups to support and develop new drone delivery services, which is favourable to new drone businesses. Moreover, the Government wants to further develop new drone delivery services in the Greater Bay Area that offer new cross-border business opportunities. There are strong and positive political influences to support LAE and the development of a new drone delivery business for FSL.

Economic influences

Economic influences involve economic trends and activities that impact the company and business. LAE is an emerging economy that offers new economic drivers to drive a slow economy and offer new business opportunities to FSL to develop a new drone delivery service business. Such a new drone delivery business offers new upstream and midstream business activities to the Hong Kong economy and offers opportunities to FSL. FSL mainly benefits from the new downstream business activities to offer new drone delivery services to customers in Hong Kong. The LAE of new drone delivery services will speed up economic recovery and the growth of the Hong Kong economy, thus offering business opportunities for FSL.

Social influences

Social influences include social attitudes toward a company and business. The public in Hong Kong have accepted drone demonstration shows and business services that offer strong and positive social influence and support new drone delivery services in Hong Kong. The growth of drone education and public interest, including operating drones for leisure activities, have also increased social acceptance of new drone delivery services. Nevertheless, there are public concerns with regards to safety, privacy, personal data security, and breach issues, all of which may affect the development of new drone delivery services. Thus, the new business poses both opportunities and threats to FSL.

Technological influences
Technological influences examine how technological changes affect a company and business. LAE involves new aviation infrastructure, driving systems, intelligent connectivity technology, automation, and artificial intelligence, all of which are new and changing rapidly and pose strong threats to the success of LAE and new drone delivery services, especially if these technologies have not been fully developed and successfully implemented. These could affect the timing of when FSL can start to operate the new drone delivery service business. Moreover, the lack of pre-existing low altitude airway networks and routes, air travel control, and drone parking all pose negative technological threats to FSL's launching of its new drone delivery service business. FSL is the user of these new technologies, and the speed and success to develop these new technologies determines the success of the new drone delivery business, which poses uncertainty and threats to FSL.

Environmental influences
Environmental influences take into account environmental considerations that impact a business and company. There are strong and positive environmental influences on LAE and the new drone delivery service business due to the industry's low carbon emissions and low air pollution, which reduce carbon footprints in the environment. Nevertheless, there are also strong and negative environmental influences concerning noise pollution and pollution from the disposal of drones and batteries. Drone malfunctions may also cause hill fires, which create losses to the public. Unless there are new laws to control noise pollution, environmental pollution, disposal of drones and batteries, and technical specifications of drones to avoid accidents, these negative influences will be obstacles to the development of LAE and the new drone delivery business. This poses uncertainty and threats to FSL.

Legal influences
Legal influences cover legal considerations that businesses may need to consider and comply with when operating businesses in particular jurisdictions. There are strong and negative influences to LAE and the new drone delivery business due to new unsolved LAE legal issues regarding privacy, data security, product liabilities, and civil aviation. Existing laws and regulations must also be updated, and new laws must be issued to protect the interests of the public surrounding LAE. The uncertainty in legal influences poses strong threats to FSL.

In conclusion, there are strong and positive opportunities regarding economic and political aspects if FSL were to develop a new drone delivery service LAE business. On the other hand, there are mixed positive and negative influences from social, technological, environmental and legal aspects. To be the forerunner in the drone delivery service business, FSL should start planning its new drone delivery service business while working closely with the government and industry stakeholders to resolve the negative influences highlighted above.

Answer 2

Since LAE has a significant impact to the environment, a Sustainability Balanced Scorecard is more appropriate to measure the performance of FSL's drone demonstration show business. A Sustainability Balanced Scorecard that incorporates financial, non-financial, and sustainability measurements and that covers all relevant areas of performance from an objective and unbiased perspective should be proposed to FSL's board of directors. There are five elements in a Sustainability Balanced Scorecard, including financial, customer, internal processes, learning and growth, and sustainability.

Financial performance measurements focus on how a company performs in terms of return to shareholders and whether financial objectives are being achieved. Since FSL is a subsidiary of DGL, it is an investment centre and thus it should use gross profit ratio, net profit ratio, and return on investment to measure the profitability of business financial performance. For liquidity, since FSL has insignificant amounts of inventory and all sales and purchases are made on credit, FSL can use either current ratio or quick ratio, days of receivables, and days of payable to measure the liquidity performance of the company. Moreover, FSL only has limited amounts of overdraft facilities, so it may use less effort to measure gearing.

Customer performance measurements are designed to determine what a company must do to satisfy its customers and achieve its financial objectives. FSL has difficulty retaining customers, so FSL should use customer churn ratio to measure customer retention. FSL can also use customer satisfaction surveys to find the reasons for losing customers and to measure customer satisfaction. FSL can also use the customer profitability index to prioritise serving profitable customers to improve profitability.

Internal processes focus on identifying and measuring what a company must excel at in order to successfully support its service and marketing strategies to achieve its financial objectives. FSL can use drone defective ratio to measure drone failure performance and find reasons for the cancelation of drone demonstration shows. FSL also needs to keep maintenance and system failure records of the global position systems to improve the system and to avoid cancelations of shows and drone accidents during demonstration shows. Records of drone flying time and battery utilisation can also help with planning future drone demonstration shows more economically, productively, and accurately.

Learning and growth are designed to ascertain what infrastructure and knowledge base a company must establish, build, and maintain to create long term growth and improvements. Due to the repetitiveness and overlapping nature of drone demonstration shows, FSL can use customer surveys to measure public opinion on the innovation and creativity of its drone demonstration show performance to learn how FSL can improve its shows. FSL should also design more innovative and creative shows in a timely fashion to attract tourists and customers. Moreover, FSL should further develop new climate monitoring information systems to monitor changes of weather and to inform the public as soon as possible regarding any possible show cancelations. FSL should also set milestones and timelines regarding when it plans to start the new drone delivery service business to ensure timely delivery of the drone demonstration shows.

Sustainability measures the environment, social, and ethical performance of a business. FSL can measure the numbers regarding disposal of drones and batteries to assess the impact on the environment and to set up environmental policies to reduce said impact. FSL can also monitor carbon emissions to measure the carbon footprint of each show and to formulate policy to minimise carbon emissions at each show. By working with drone manufacturers, FSL can also measure decibels generated by its drones during drone demonstration shows so that it can set up key performance indicators regarding noise pollution caused by drones.

Answer 3

As a professional accountant working in the commercial sector, the finance manager should comply with the Code of Ethics for Professional Accountants -- particularly Part 2 Section 200 to 270.

Pressure to breach fundamental ethical principles
A professional accountant may be deterred from acting objectively because of pressure and attempts to exercise undue influence over the accountant regarding compliance with ethical principles. The finance manager is under pressure to prepare the proposal within a week, which may not be possible to accomplish. The finance manager should thus inform the financial controller that he needs more time to work on the proposal. The finance manager should also prepare a realistic forecast with realistic assumptions without considering the job probation issues.

Acting with sufficient expertise in preparation and presentation of financial information
Acting without sufficient expertise creates a threat of self-interest wherein financial or other interest may inappropriately and negatively influence a professional accountant's judgement or behaviour to comply with the principle of professional competence and due care. The finance manager has been working in the finance department of a local bank for 10 years and does not have sufficient expertise to prepare a financial forecast for a commercial company in an emerging economy. Moreover, this is the first time the finance manager has prepared an ITF application proposal, indicating that he has no experience with it. The finance manager should thus talk to the financial controller to hire an external professional consultant with relevant LAE and drone business knowledge and skills to assist with preparation of the proposal.

Inducements and responding to non-compliance with law and regulations
Offering inducements creates a familiarity threat, which is a threat due to a long or close relationship with a third party wherein professional accountants are not able to comply with principles of integrity, objectivity, and professional behaviour. It also creates a threat to non-compliance with the principle of professional behaviour, resulting in violation of relevant laws and regulations and discrediting the reputation of the profession. The finance manager has offered a luxury dinner and has given gifts to his nephew working in ITF to get "insider" information to aid in preparing the proposal. Due to offering an advantage to another as an inducement for benefits, this is a breach of the Prevention of Bribery Ordinance (Cap. 201). Particularly, the finance manager has offered advantages to a government officer to obtain benefits that are specifically prohibited by the Ordinance. The finance manager should not offer inducements to his nephew to obtain "insider" information to aid in preparing the proposal.

Conflict of interest
A conflict of interest creates threats to self-interest and prevents compliance with the principle of objectivity, such as when a professional accountant compromises professional and business judgement due to undue influence from others. The finance manager is facing a conflict of interest in the preparation of a profitable forecast due to undue influence from the financial controller, who has linked the finance manager's job probation to the success of obtaining funding from ITF. The finance manager should try his best to prepare the proposal with integrity, objectivity, and professional competence and due care and should not forsake the Code of Ethics for his personal interest (i.e. job probation), because any breaches of the Code of Ethics will affect the future career of the finance manager.

Answer 4(a)

Plan A:

Equity beta = 0.8 x (1 + (0.2/0.8) x (1 – 0.165)) = 0.967

The cost of equity is:
= 0.03 + 0.967 x 0.05
= 0.0784 or 7.84%

Plan B:

Equity beta = 0.8 x (1 + (0.5/0.5) x (1 – 0.165)) = 1.468

The cost of equity is:
= 0.03 + 1.468 x 0.05
= 0.1034 or 10.34%

Comparing Plan A and Plan B, FSL should choose Plan A because it has a lower cost of equity. It is noted that when companies make financing decisions, the Weighted Average Cost of Capital (WACC) is normally used to determine the optimal capital structure. However, in this case, only the cost of equity is considered because the cost of debt is unusually higher than the cost of equity, making debt financing unfavorable. As debt increases, the leveraged equity beta and required return of equity also increase. Therefore, FSL must ensure the return on investment is equal to or above the required return on equity, otherwise FSL may not have sufficient earnings to meet shareholders' return requirements.

Answer 4(b)

Year	0	1	2	3	
Earnings					
EPS growth rate (%)		20	30	8.27	(Note 1)
EPS (HK$)	2	2.4	3.12	3.38	
Dividends					
EPS (HK$)	2	2.4	3.12	3.38	
Dividend payout rate (%)		0	20	20	
Dividend (HK$)		0	0.62	0.68	

The perpetuity value of the dividend is:
= 0.68 / (0.1034 – 0.0827)
= HK$32.85

The current share price should be:
= 0.62 / $(1+0.1034)^2$ + 32.85 / $(1+0.1034)^2$
= HK$27.49

Note 1:
Sustainable growth rate:
= (1 – 20%) x 10.34%
= 8.27%

An estimated annual EPS growth rate of 30% exceeds the expected return on equity under Plan B from Year 2 onwards, implying an infinite share price. A sustainable growth rate is therefore required, and in this case it is estimated using Gordon's growth rate approximation. Other well-justified estimates may also be applied.

* * * END OF SECTION A * * *

SECTION B – ESSAY / SHORT QUESTIONS (Total: 50 marks)

Answer 5(a)

Insufficient involvement by LLD management:

Although a competent due diligence team was present to support the merger or acquisition of SEAA, it is also essential for LLD's management to be actively engaged in each stage of the process. It is possible that restrictions during COVID-19 limited the ability of LLD's management to dedicate sufficient time and effort to the acquisition, e.g. on-site visits, which could have led to the failure of the acquisition.

Under- or over-valuation:

The assets and value listed during a merger may not always accurately represent reality, especially during COVID-19 when SEAA suffered from travel restrictions and was under financial difficulties. Such inaccurate numbers could lead to significant disruptions in the acquisition process and provide an incorrect basis for LLD to make financial forecasts and assumptions, potentially leading to wrong strategic decisions and financial instability.

Business integration related issues:

Integrating two business operations can be challenging. The acquisition may fail if LLD does not effectively integrate their systems and business processes, such as ticket ordering and customer relationship management, as well as staff recruitment and training, and aircraft repair and maintenance in particular depend on the age of the plane. If LLD does not have much acquisition experience, then the benefit of business synergy between the two companies may not materialise, which could lead to operational inefficiencies or even financial losses.

Corporate culture issues:

Cultural issues are common in multinational acquisitions. Understanding the corporate cultures of both SEAA and LLD is crucial for the success of the acquisition. LLD may lack sufficient knowledge in areas such as legal and regulatory environments, local customer preference and consumer behaviour, and how SEAA's customers contribute to market development and synergy. Furthermore, it is important for LLD to have the capability to understand Malaysian culture in order to effectively manage staff and meet customer requests.

Underestimating capacity and manpower requirements:

Although LLD conducted comprehensive due diligence when evaluating whether SEAA could manage the additional demands effectively, the capacity of an airline business is heavily dependent on various factors such as the number of aircraft, pilots, and approved routes. The demand for pilots as well as aircraft repair and maintenance services have increased significantly, especially after the resumption of international travel post-COVID-19. As a result, SEAA's capacity would be substantially impacted if it cannot secure enough pilots or repair and maintenance services.

Higher than expected operating costs:

When acquiring a distressed target business, it is very common for the acquirer to pay more than expected in order to see a profit. SEAA was under financial distress during COVID-19, and it may have significant structural problems and require extensive restructuring, additional capital infusion, or other costly interventions to fix those problems before it becomes profitable. At the same time, future growth prospects, especially surrounding the rebound of travel after COVID-19, may be overly optimistic. Thus, LLD may be required to invest significant resources to help SEAA's financial health recover.

Answer 5(b)

Income-based approaches:

This approach includes dividend discount models, earnings-based models and residual income models, and free cash flow models, which convert future amounts (e.g., income, expenses, and cash flows) to a discounted valuation that reflects current market expectations about future earnings, dividends, or cash flows. Additionally, earnings-based models are relatively simple approaches to estimating valuation by applying market or specific price-earnings ratio (i.e. equity multiples) and relating the value of an entity directly to its earnings/ cashflow. However, considering the losses and negative cash flow position of SEAA, this model is not applicable.

Market-based approaches:

This comprises the use of observed market prices and comparable company valuations, along with recent transaction data, for valuing SEAA. This approach does not take into account income or cash flow; instead, it relies on the valuation of similar airlines that have comparable asset sizes and financial performance. As such, this is suitable for the case of SEAA, who do not generate income and positive cash flow. By examining these comparable companies and transactions, LLD can estimate the value of SEAA and provide a realistic and market-based perspective to a potential buyer.

Asset-based approaches:

Asset-based approaches value an entity by valuing the entity's assets and liabilities to identify the company's net asset value.

In this case, since SEAA is not generating any profits nor positive cash flow, traditional income-based valuation methods, which rely on projecting future income, expenses, and cash flows to determine the present value, are not applicable.

Conclusion:

It would be more appropriate to employ market or asset-based valuation methods that can provide a realistic assessment of SEAA's worth based on factors other than current profitability. These methods focus on the company's tangible assets, such as SEAA's numerous aircraft which it owns, which are valuable assets and critical to the operations of an airline company. These are particularly pertinent when SEAA is not currently profitable, as it relies on the intrinsic value of the company's physical assets rather than its earnings. By doing so, LLD can provide a realistic assessment of SEAA's worth to potential buyers, ensuring that it has a comprehensive understanding of the company's value despite its lack of current profitability.

Answer 6(a)

	Year 0	Year 1	Year 2	Year 3	Year 4
	RMB millions	RMB millions	RMB millions	RMB millions	RMB millions
Leasing saving			16.5	18.2	20.0
Renovation	(5.56)				
Buying of hardware		(37)			
Operating cost			(7)	(7)	(7)
Selling of hardware					18
Net cash flow	(5.56)	(37)	9.5	11.2	31
Discounted factor	1	0.935	0.873	0.816	0.763
Present value	(5.56)	(34.60)	8.30	9.14	23.65

The net present value is RMB0.93 million.

Since the NPV is positive, the IRR is expected to be higher than the cost of capital of LCT's 7%. By trial and error with the closest integer, i.e. 7% and 8%, we get zero when we apply 8%, so IRR is 8%.

Since the NPV is positive and the IRR is higher than the cost of capital of LCT, LCT should accept this project.

Answer 6(b)

With an NPV of RMB0.93 million, the recommendation would change when NPV is reduced to below zero.

RMB0.93 million / RMB5.56 million = 16.73%

An increase in renovation cost at 16.73% would reduce the Net Present Value by RMB0.93 million and change the recommendation provided in Part (a).

Answer 6(c)

6-year forward contract rate = 7.2 x $(1 + 1.5\%)^6$ / $(1 + 3\%)^6$
= 7.2 x 0.9157
= 6.59

The theoretical 6-year forward contract rate of USD / RMB is 6.59.

Answer 7(a)

Carbon emissions (kg)
= Number of trucks x Kilometres travelled per truck in the year / Average kilometre per litre of diesel x Carbon emission per litre of diesel fuel

2024: 38 x 16,000 km / 7 km per litre x 2,700 kg/L
= 234,514,285.71 kg

2025: 41 x 18,000 km / 5 km per litre x 2,600 kg/L
= 383,760,000.00 kg

2024 Carbon credits to pay
= Carbon emissions (kg) / 907 x Carbon credit (per ton)
= 234,514,286 / 907 x HK$50
= HK$12,928,020

2025 Carbon credits to pay
= 383,760,000 / 907 x HK$50
= HK$21,155,458

Answer 7(b)

There are numerous examples of greenwashing behaviours, such as those discussed in Vollero's 2022 book "Greenwashing: Foundations and Emerging Research on Corporate Sustainability and Deceptive Communication". Shadow's proposal includes the following points, which feature strong similarities to examples of greenwashing given in Vollero's book.

i) Shadow's reporting practices have changed since last year due to neglecting carbon emissions generated from its main business processes. This has led to insufficient, incomplete, or inconsistent reporting of Shadow's sustainability performance. The omission of material information regarding carbon emissions from key business processes is a typical example of greenwashing due to the lack of disclosure of essential data.

By disclosing only the carbon emissions of the electric vehicles used by its management, which represents selective disclosure, and by presenting diverging reports of material data from key business processes to less relevant data, such selective disclosure is also considered greenwashing.

ii) Reducing the scale of tree planting activities is considered immaterial virtue signalling. By limiting the invitation of tree planting activities to only seven board members, the initiative constitutes an insignificant component of the overall environmental footprint. This fails to involve a broader participation of the company and thus diminishes the impact. Reporting tree planting activities with only a very small participation by Shadow also overemphasises Shadow's environmental efforts and provides a misleading impression to the public, which is an example of greenwashing.

iii) The environmental certificate on Shadow's website did not provide details regarding the certification process, accreditation framework, or other relevant information, and the process was not certified by a qualified party. This approach demonstrates poorly defined and obscure metrics and uses vague methodologies and measurement techniques that lack clear definitions. Such practices are misleading, as they fail to provide a transparent and accurate representation of Shadow's environmental impact. This is considered insufficient, incomplete, or inconsistent external validation of ESG reporting and ultimately constitutes a form of greenwashing.

* * * END OF EXAMINATION PAPER * * *